Content Monetization Modeler: YouTube Ad Revenue Prediction Using Machine Learning

1. Project Overview

The objective of this project is to develop a Machine Learning model capable of predicting YouTube ad revenue based on video performance metrics and audience engagement data. Accurate revenue prediction helps content creators and businesses understand the factors influencing monetization and make data-driven decisions to improve earnings.

The project includes data preprocessing, feature engineering, model training, evaluation, and deployment through an interactive Streamlit application.

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2. Problem Statement

YouTube creators generate revenue through advertisements displayed on their videos. Ad revenue depends on several factors such as views, watch time, audience engagement, device type, and geographical location. The goal of this project is to build a predictive model that estimates ad revenue based on these features.

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3. Dataset Description

The dataset contains information related to YouTube video performance and audience engagement.

Features Used

- Views
- Likes
- Comments
- Watch Time Minutes
- Video Length Minutes
- Subscribers
- Category
- Device Type
- Country
- Year
- Month

- Day

Target Variable

- Ad Revenue (USD)

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4. Data Preprocessing

The following preprocessing steps were performed:

1. Removed unnecessary columns such as Video ID.
2. Handled missing values.
3. Extracted date-related features such as Year, Month, and Day.
4. Applied One-Hot Encoding to categorical variables:
    - Category
    - Device
    - Country
    - Month
5. Scaled numerical features using StandardScaler.
6. Split the dataset into Training and Testing datasets.

After preprocessing, the final dataset contained 32 input features.

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5. Feature Engineering

Feature engineering was performed to improve model performance. Categorical variables were transformed into numerical format using One-Hot Encoding. Date information was decomposed into separate components including year, month, and day.

These engineered features enabled machine learning algorithms to effectively learn patterns affecting ad revenue.

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6. Machine Learning Models Implemented

Five regression models were trained and evaluated:

1. Linear Regression

A statistical regression model used to establish a linear relationship between input features and ad revenue.

2. Decision Tree Regressor

A tree-based model that learns decision rules from the training data.

3. Random Forest Regressor

An ensemble learning technique that combines multiple decision trees to improve prediction accuracy.

4. Gradient Boosting Regressor

A boosting algorithm that sequentially improves prediction performance by correcting previous errors.

5. K-Nearest Neighbours (KNN) Regressor

A distance-based algorithm that predicts values using neighbouring data points.

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7. Model Evaluation Metrics

The following evaluation metrics were used:

R^2 Score

Measures how well the model explains the variance in the target variable.

Mean Absolute Error (MAE)

Measures the average absolute difference between actual and predicted values.

Root Mean Squared Error (RMSE)

Measures the standard deviation of prediction errors.

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8. Model Performance Comparison

Model	R^2 Score	MAE	RMSE
Linear Regression	0.9519	3.17	13.60
Gradient Boosting Regressor	0.9517	3.69	13.64
Random Forest Regressor	0.9510	3.99	13.74
Decision Tree Regressor	0.8921	5.79	20.38

Model	R^2 Score	MAE	RMSE
KNN Regressor	0.6622	29.06	36.06

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9. Best Model Selection

Based on the evaluation results, Linear Regression achieved the highest R^2 Score and the lowest prediction errors.

Best Model

- Model: Linear Regression
- R^2 Score: 0.9519
- MAE: 3.17
- RMSE: 13.60

The model explains approximately 95% of the variation in ad revenue, making it highly effective for revenue prediction.

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10. Feature Importance Analysis

The most influential features affecting ad revenue were:

Feature	Coefficient
Watch Time Minutes	59.63
Likes	8.65
Comments	2.11
Views	0.64
Year	0.21

Key Insights

1. Watch Time Minutes is the strongest factor influencing ad revenue.
2. Videos with higher engagement (likes and comments) tend to generate more revenue.
3. Higher view counts positively impact earnings.

4. Seasonal effects were observed in months such as September, November, and December.
5. Audience engagement is more important than views alone for maximizing revenue.

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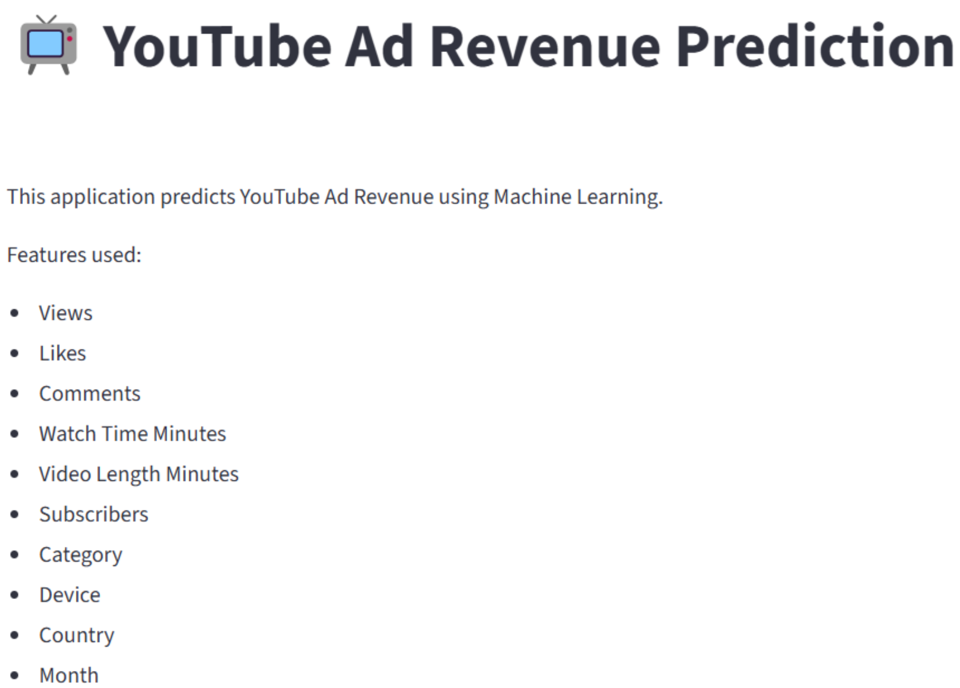
11. Streamlit Application Development

A Streamlit web application was developed to allow users to interactively predict YouTube ad revenue.

Features of the Application

- User-friendly interface
- Sidebar navigation
- Revenue prediction functionality
- Real-time predictions
- Interactive input fields
- Display of predicted ad revenue

The application loads the trained Linear Regression model and scaler using Pickle files and generates predictions based on user input.



Model Performance:

- R^2 Score: 0.9519
- MAE: 3.17
- RMSE: 13.60

Best Model: Linear Regression

💰 Predict YouTube Ad Revenue

Views

10000 − +

Subscribers

100000 − +

Likes

1000 − +

Category

Education

Comments

100 − +

Device

Desktop

Watch Time Minutes

5000.00 − +

Country

AU

Video Length Minutes

10.00 − +

Month

April

Year

2025 − +

Day

1 − +

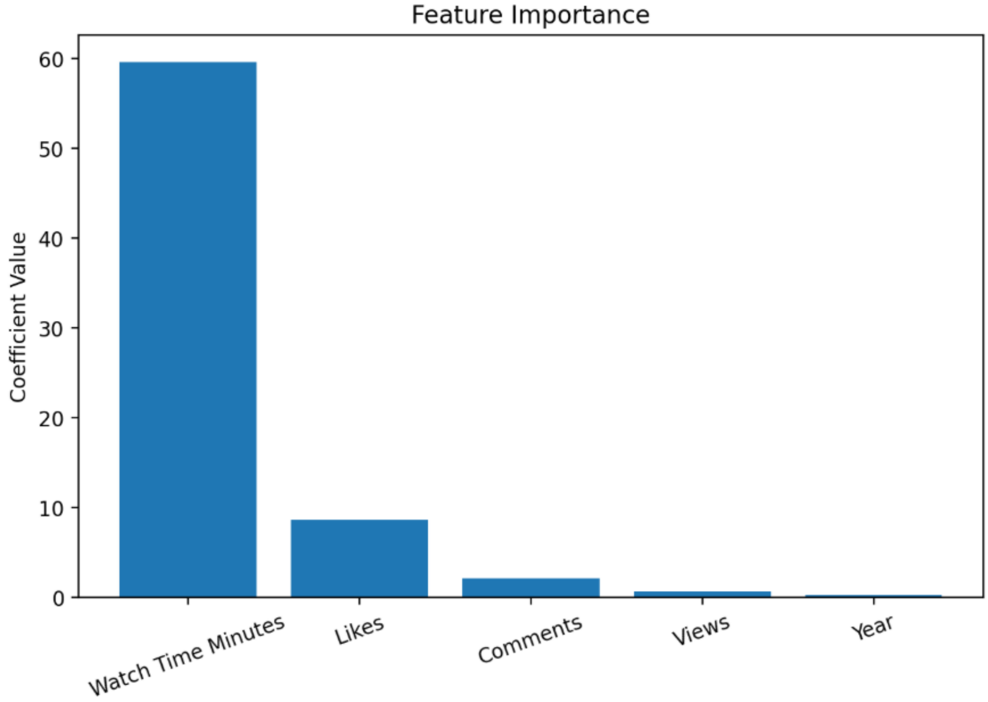
Predict Revenue

💰 Predicted Ad Revenue: $94.82

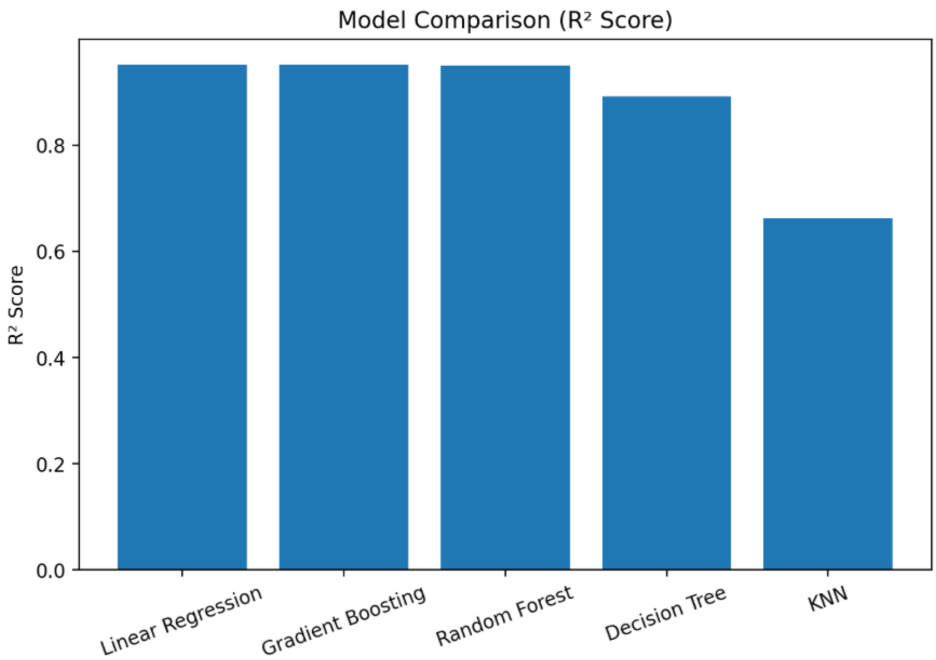
Input Summary

	views	likes	comments	watch_time_minutes	video_length_minutes	subscribers	year	day
0	10000	1000	100	5000	10	100000	2025	1

Top Features Influencing Ad Revenue

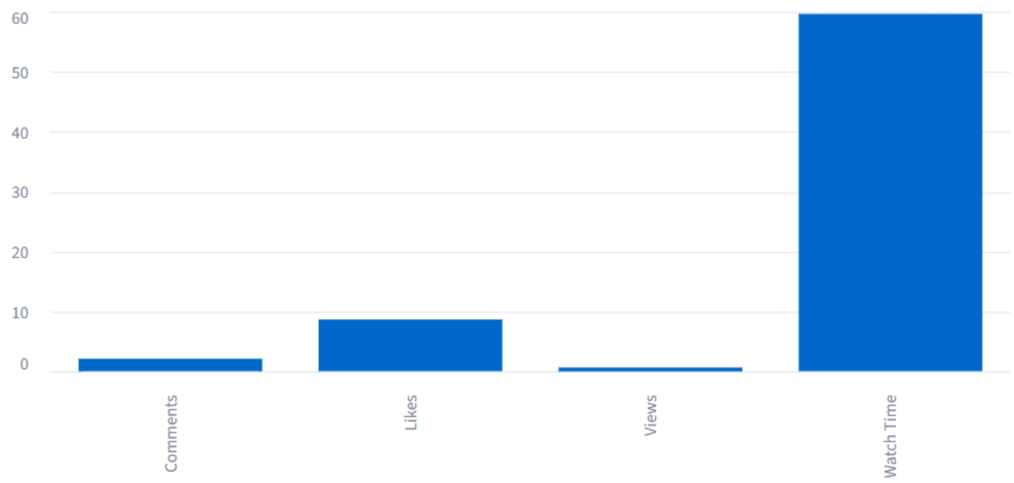


Regression Model Performance

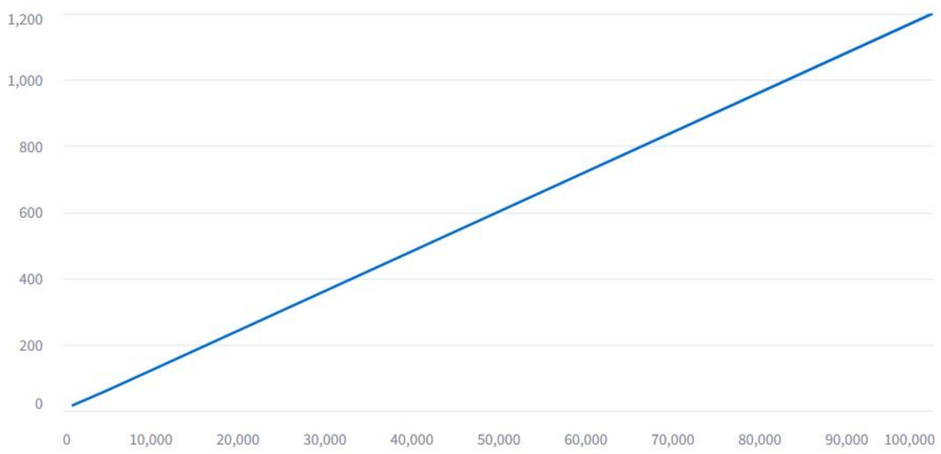


Revenue Driver Analysis

Revenue Driver Analysis



Revenue Growth Trend



Best Model Performance

R^2 Score	MAE	RMSE
0.9519	3.17	13.60

Linear Regression was selected as the final model because it achieved the highest R^2 Score and lowest prediction errors.

📌 Key Insights

- Watch Time Minutes is the strongest predictor of ad revenue.
- Likes and Comments significantly improve monetization.
- Videos with higher engagement tend to generate more revenue.
- Linear Regression achieved the best overall performance with an R^2 Score of 95.19%.
- Audience engagement is more influential than views alone.

12.Project Structure

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CONTENT MONETIZATION MODELER PROJECT/
│
├── Report/
│   └── Content Monetization Modeler Report.pdf
│
├── Screenshots/
│   ├── Home Page.png
│   ├── Prediction Page.png
│   ├── Prediction Result.png
│   └── Visualizations Page.png
│
├── model.py
├── monetization analysis.ipynb
├── README.md
```

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├── requirements.txt
├── scaler.pkl
├── youtube_ad_revenue_dataset.csv
└── youtube_ad_revenue_model.pkl
```

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13. Results

The project successfully achieved the following objectives:

- Cleaned and preprocessed the dataset.
- Built and evaluated five regression models.
- Selected the best-performing model.
- Identified key drivers of YouTube ad revenue.
- Developed an interactive Streamlit application.
- Generated accurate revenue predictions with an R^2 Score of 95.19%.

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14. Conclusion

This project demonstrates the successful application of Machine Learning techniques for predicting YouTube ad revenue. Through data preprocessing, feature engineering, and model evaluation, Linear Regression was identified as the most effective model. The analysis revealed that watch time, likes, comments, and views are the most significant factors influencing revenue generation.

The deployed Streamlit application provides a practical tool for content creators to estimate potential earnings and understand the key factors affecting monetization.

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15. Future Scope

1. Integrate real-time YouTube Analytics data.
2. Implement advanced models such as XGBoost and LightGBM.
3. Deploy the application on cloud platforms.
4. Add revenue forecasting capabilities.
5. Develop personalized recommendations for content creators to maximize earnings.